

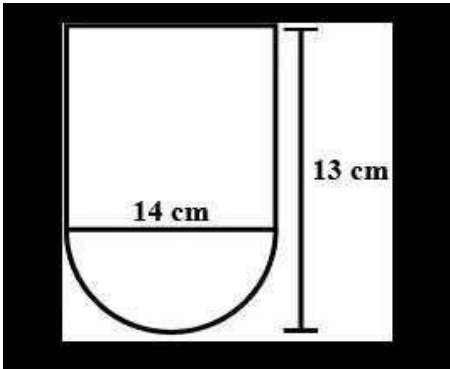
	Pre-Board Examination (2024-25) Marking Scheme Class- X Subject- Mathematics Basic (241) Time: 3 Hours MM:80	
Q.No	Answers	Marks
	Section A	
1	b) 4	1
2	b) 12	1
3	c) Infinitely many solutions	1
4	c) No real roots	1
5	b) 10	1
6	d) (-7,0)	1
7	c) 1	1
8	a) $\cos^2 A$	1
9	b) 45°	1
10	c) $ac = bx$	1
11	c) 2	1
12	b) 50°	1
13	b) 462 cm^2	1
14	a) 314 cm^2	1
15	d) 216 cm^3	1
16	d) 13	1
17	$(c) \text{ Mean} = (3\text{Median} - \text{Mode})/2$	1
18	a) $1/8$	1
19	d) Assertion (A) is false but reason (R) is true.	1
20	a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).	1
	Section B	
21	Adding eq we get $x + y = 1$ Subtracting eq we get $x - y = 3$ Solving $x = 2, y = -1$ OR $kx + y = k^2$ and $x + ky = 1$	$\frac{1}{2}$ $\frac{1}{2}$

	$\frac{k}{1} = \frac{1}{k} = k^2$ $K^2 = 1$ $k = \pm 1$	$\frac{1}{2}$ $\frac{1}{2}$
22	For correct proof	2
23	Since the triangles are similar therefore $\frac{8}{10} = \frac{DE}{6}$ $DE = 4.8 \text{ cm}$	1 1
24	For given and to prove Congruency of triangles For CPST	$\frac{1}{2}$ 1 $\frac{1}{2}$
25	Using the formula Area of minor segment = $\frac{\theta}{360} \times \pi r^2 - \frac{1}{2} \text{ side} \times \text{side}$ $= 28.50 \text{ cm}^2$ OR Area swept by both the wipers = $\frac{2 \times \theta}{360} \times \pi r^2$ $= 924 \text{ cm}^2$	1 1 1 1
SECTION C		
26	For correct proof	3
27	Solution: We have, $2x^2 - 5x - 3 = 0$ $= 2x^2 - 6x + x - 3$ $= 2x(x - 3) + 1(x - 3)$ $= (x - 3)(2x + 1)$ Zeroes are: $x - 3 = 0$ or $2x + 1 = 0$ $\Rightarrow x = 3$ or $x = -\frac{1}{2}$ Since the zeroes of required polynomial is double of given polynomial Zeroes of the required polynomial are: $3 \times 2, (-\frac{1}{2} \times 2)$, i.e., 6, -1 Sum of zeroes, $S = 6 + (-1) = 5$ Product of zeroes, $P = 6 \times (-1) = -6$ Quadratic polynomial is $x^2 - Sx + P$ $\Rightarrow x^2 - 5x - 6 \dots (i)$ Comparing (i) with $x^2 + px + q$ $p = -5, q = -6$	$1 \frac{1}{2}$ $\frac{1}{2}$ 1
28	The given equations are $2x + ky = 11 \dots (i)$ $x - 2y = -12 \dots (ii)$ We know that, The system of linear equations is in	1

	the form of $A_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ Compare (i) and (ii), we get $a_1 = 2$, $b_1 = k$ and $c_1 = -11$ $a_2 = 1$, $b_2 = -2$ and $c_2 = 12$ The equations has a unique solution $a_1/a_2 \neq b_1/b_2$ If $2/1 \neq k/-2$ $k \neq -4$ Thus k can take any values except -4.	1 1
29	We have, $(\operatorname{cosec}\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$ LHS = $(\operatorname{cosec}\theta - \cot\theta)^2$ $= (1/\sin\theta - \cos\theta/\sin\theta)^2$ $= (1 - \cos\theta)^2/\sin^2\theta$ $= (1 - \cos\theta)^2/1 - \cos^2\theta$ $= (1 - \cos\theta)^2/(1 - \cos\theta)(1 + \cos\theta)$ $= (1 - \cos\theta)/(1 + \cos\theta)$ OR $\frac{\cos A + 1 + \sin A}{1 + \sin A} = 2 \sec A$ LHS = $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A}$ $= \frac{\cos^2 A + 1 + \sin^2 A + 2 \sin A}{(1 + \sin A)(\cos A)}$ $= \frac{2 + 2 \sin A}{(1 + \sin A)(\cos A)}$ $= \frac{2(1 + \sin A)}{(1 + \sin A) \cos A}$ $= 2 \sec A$	1 1 1 1 1 1
30	In the given figure, O is the centre of a circle. PT and PQ are tangents to the circle from an external point P. Join OT and OQ. If $\angle TPQ = 70^\circ$, find $\angle TRQ$. We know that the radius and tangent are perpendicular at their point of contact. $\angle OTP = \angle OQP = 90^\circ$, Now, In quadrilateral OQPT $\angle QOT + \angle OTP + \angle OQP + \angle TPQ = 360^\circ$ (angle sum property of quadrilateral) $\angle QOT + 90^\circ + 90^\circ + 70^\circ = 360^\circ$ $\angle QOT + 250^\circ = 360^\circ$ $\angle QOT = 360^\circ - 250^\circ$ $\angle QOT = 110^\circ$ We know that the angle subtended by an arc at the centre is double of the angle subtended by the arc at any point on the circumference of the circle. $\angle TRQ = \frac{1}{2} \angle QOT \Rightarrow \angle TRQ = \frac{1}{2} \times 110 = 55$ OR We have, $OP = OQ = 5\text{cm}$ (radii of larger circle) And $QR = 3\text{ cm}$	1 1 1 1 $\frac{1}{2}$

	Since PQ is tangent to the smaller circle Therefore $OR \perp PQ$ (By theorem) In $\triangle OPR$ AND $\triangle OQR$ $\angle ORP = \angle ORQ$ (Each 90°) $OR = OR$ (Common) $OP = OQ$ (radii of same circle) Therefore $\triangle OPR \cong \triangle OQR$ (RHS congruence) $\Rightarrow PR = RQ$ [CPCT] In $\triangle OPR$, $PR^2 = OP^2 - OR^2$ $= (5)^2 - (3)^2 = 16\text{cm}$ $\Rightarrow PR = 4\text{cm}$ Therefore $PQ = 2PR = 2 \times 4 = 8\text{cm}$	$\frac{1}{2}$ 1
31	There are 6 red face cards. These are removed. Thus, remaining number of card $= 52 - 6 = 46$. i. Number of red cards now $= 26 - 6 = 20$. Therefore, $P(\text{getting a red card}) = (\text{Number of favourable outcomes}) / (\text{Total number of possible outcomes}) = 20/46 = 10/23$ ii. Number of face cards now $= 12 - 6 = 6$. Therefore, $P(\text{getting a face card}) = (\text{Number of favourable outcomes}) / (\text{Total number of possible outcomes}) = 6/46 = 3/23$ iii. The number of card of clubs $= 13$. Therefore, $P(\text{getting a card of clubs}) = (\text{Number of favourable outcomes}) / (\text{Total number of possible outcomes}) = 13/46$ Thus, the probability that the drawn card is a card of clubs is $13/46$	1 1 1
	SECTION D	

32	Distance travelled by train = 480 km	1
	Let the speed of train = x km/h	
	Time taken = 480/x	1
	If the speed of train = (x - 8) km/h	
	Time taken = 480 / x-8	
	$x^2 - 8x - 1280 = 0$	1
	x = 40 and x = -32 (Neglect)	2
	so, speed of train = 40 km/h.	
	OR	
	Let the speed of the stream be x km/hr	$\frac{1}{2}$
33	So, the speed of the boat in upstream = (18 - x) km/hr & the speed of the boat in downstream = (18 + x) km/hr	$\frac{1}{2}$
	ATQ, distance upstream speed - distance downstream speed = 1	1
	$\Rightarrow \frac{24}{18-x} - \frac{24}{18+x} = 1$	2
	$\Rightarrow x^2 + 48x - 324 = 0$	1
	$\Rightarrow (x + 54)(x - 6) = 0 \Rightarrow x = -54 \text{ or } 6$ As speed to stream can never be negative, the speed of the stream is 6 km/hr	
	For correct figure given to prove	1
		1
		1
	construction	2
	correct proof	
34	Forming correct table	1 $\frac{1}{2}$
	Median = $l + \left(\frac{n-cf}{f} \right) Xh$	$\frac{1}{2}$
	Substituting correct values	1
	Substituting values and finding x=9 and y=15	2

35	Since the inner diameter of the glass = 5 cm and height = 10 cm, the apparent capacity of the glass = $\pi r^2 h$ $= 3.14 \times 2.5 \times 2.5 \times 10 \text{ cm}^3$ $= 196.25 \text{ cm}^3$ But the actual capacity of the glass is less by the volume of the hemisphere at the base of the glass. i.e., it is less by $\frac{2}{3} \pi r^3 = \frac{2}{3} \times 3.14 \times 2.5 \times 2.5 \times 2.5$ $= 32.71 \text{ cm}^3$ So, the actual capacity of the glass = apparent capacity of glass – volume of the hemisphere = $(196.25 - 32.71) \text{ cm}^3 = 163.54 \text{ cm}^3$ OR  Radius = 7 cm Height of the cylindrical portion = $13 - 7 = 6 \text{ cm}$. Area of a Curved surface of cylindrical portion is, $= 2\pi r h$ $= 2 \times \frac{22}{7} \times 7 \times 6$ $= 264 \text{ cm}^2$ Area of a curved of hemispherical portion is $= 2\pi r^2$ $= 2 \times \frac{22}{7} \times 7 \times 7$ $= 308 \text{ cm}^2$ $\therefore$ Total surface area is $= 308 + 264 = 572 \text{ cm}^2$.	$\frac{1}{2}$ 1 $1 \frac{1}{2}$ 1 1
	SECTION D	
36	(a) 6 (b) 6n (c) 60 OR 48	1 1 2
37	Yes (7,5) C(11,5); D(7,1) OR A(3,5) : B(7,9)	1 1 2

38	1. 60°	1
	2. 30°	1
	3. $16\sqrt{3}$ m	2
	OR $32\sqrt{3}$ m	